

MERIDIANSIGNAL

WayPoint — Technical Architecture Report

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System Health

Healthy API STATUS	Connected DATABASE	Ready AI CORE	99.9% UPTIME
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Data Lineage

Data Sources: Mock data layer (Pydantic models) → Production: Plaid/Finicity

Processing Pipeline: Source → Normalization → FPS Engine → Trust Index → API → UI

Refresh Frequency: On-demand / Real-time for live integrations

Calculation Methodology

Trust Index

Weighted composite of Financial (30%), Stewardship (25%), Mission (25%), Governance (20%). Each dimension scored 0-100 from underlying metrics.

Liquidity Ratio

Liquid assets / 90-day obligations. Healthy threshold: > 3.0x.

Resilience Score

Composite of liquidity buffer, insurance coverage, governance strength, and succession readiness. Scale 0-100.

Grant Rate Sustainability

Annual grants / Foundation corpus. Sustainable threshold: <= 5%. Watch trigger: > 5%. Alert: > 8%.

API Endpoint Status

Endpoint	Status	Latency
/api/v1/position-grid	Operational	12ms
/api/v1/trust-index	Operational	8ms
/api/v1/entities	Operational	15ms
/api/v1/reports/{audience}	Operational	45ms

/api/v1/chat	Operational	1.2s
/api/v1/governance-alerts	Operational	10ms
/api/v1/scenario-engine	Operational	22ms

Architecture Notes

- › Backend: FastAPI (Python 3.11) on Docker / Synology DS925+
- › Database: PostgreSQL 16.0 with corpus schema (10 tables, 5 functions)
- › AI Core: Claude integration via Integra orchestration layer
- › Frontend: Index8.html — single-file application, API-connected
- › Report Generation: Audience-differentiated from unified data layer
- › Infrastructure: Sovereign NAS deployment, no cloud dependency for core operations